

Deep Insights, Smart Moves: Advancing Traffic Prediction with Hybrid LSTM-GCN Architectures

Xinyang Zhang and Zhiying Tang

Instructor: Dr. Ren Zhao

Xi'an Jiaotong-Liverpool University, Suzhou, China

Email: Xinyang.Zhang23@student.xjtlu.edu.cn

Zhiying.Tang23@student.xjtlu.edu.cn

Ren.Zhao@xjtlu.edu.cn

2024 Theme: Revolutionizing Computing and Communication in the AI Era





CONTENT

- ❖ Background
- ❖ Data Processing
- ❖ Model
- ❖ Innovation
- ❖ Application
- ❖ Conclusion

BACKGROUND



Challenges

- Traffic congestion
- Complex road conditions
- Inconvenience to people's daily travel
- Increases the difficulty of traffic management

Limitations

- Struggle to capture the complex dynamics of actual traffic flows
- Ignore the nonlinear nature of the transportation system
- Insufficient prediction accuracy.

Dataset

- T-drive Taxi Trajectory Dataset
- Beijing_sevenpart Dataset
- PeMS04
- PeMS08

DATA PROCESSING

Data Cleaning Data Deduplication Data Integration



- Remove outliers
- Dealing with missing data
- Format standardization

dropna()

- Detect duplicates
- Remove redundant data
- Ensure data uniqueness

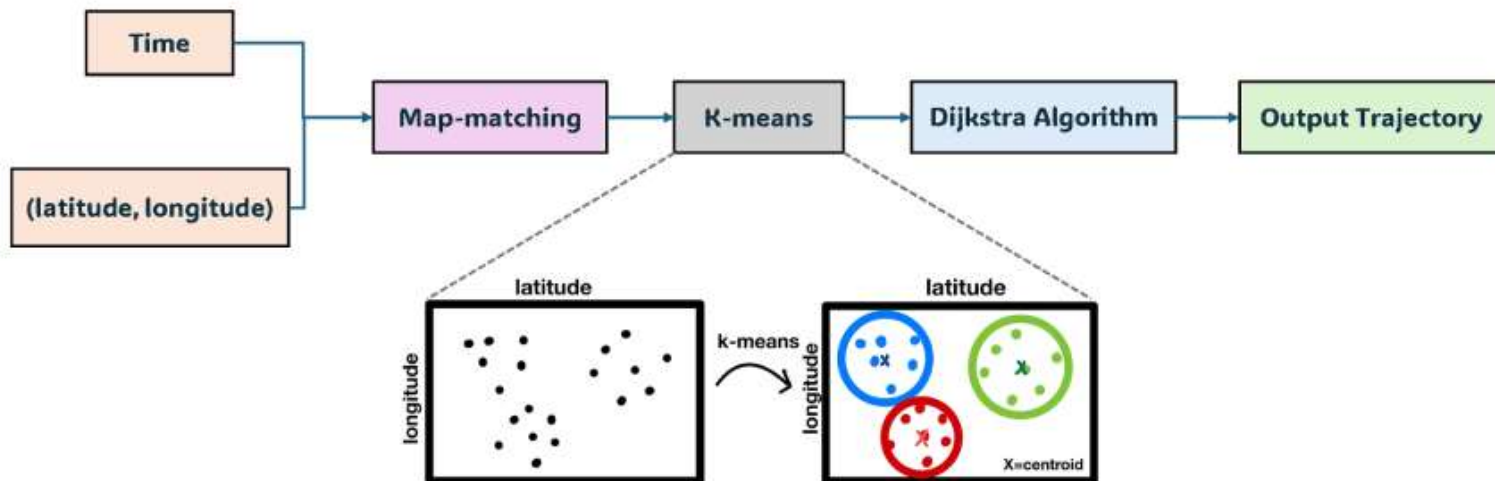
duplicated()
drop_duplicates()

- Integrate multiple data sets
- Uniform format
- Ensure data consistency

MODEL

OPTIMAL PATH PLANNING

OPTIMAL PATH PLANNING

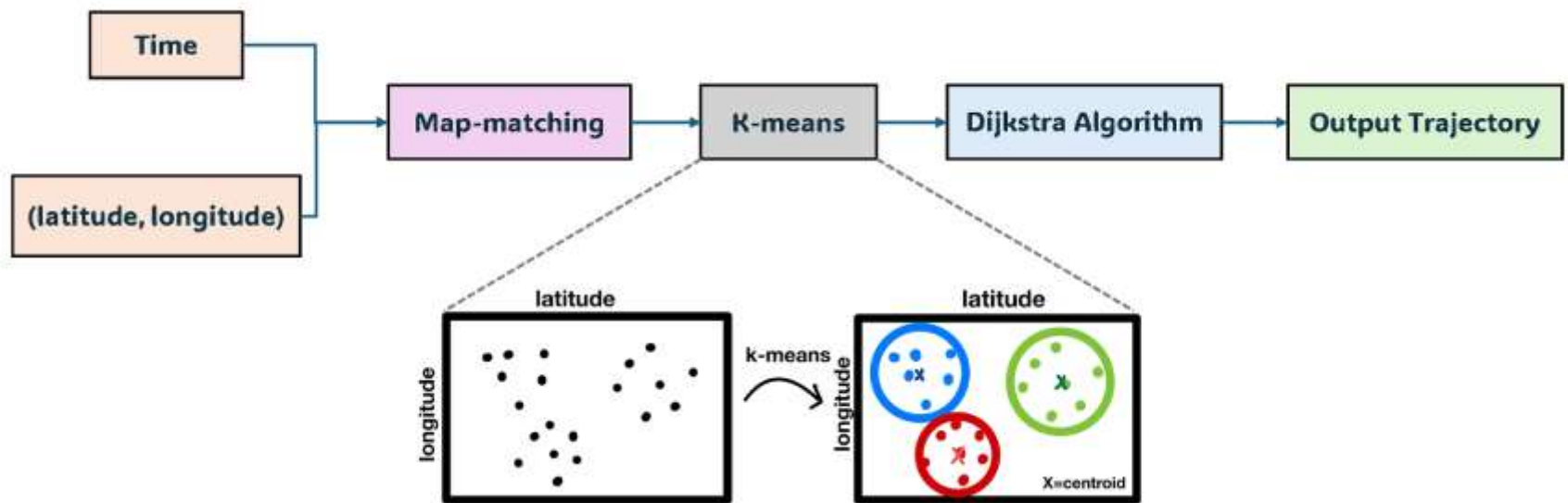


GPS
Point



Road
Network



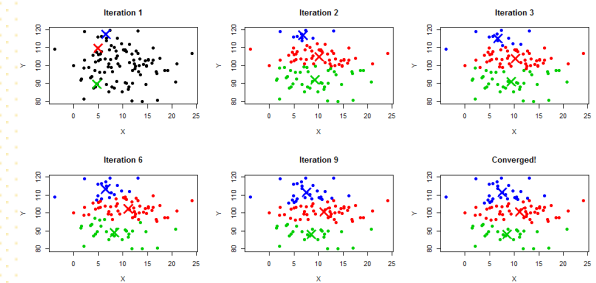


LANDMARK CALCULATION -- K-MEANS

STEP 1: Set the cluster center randomly

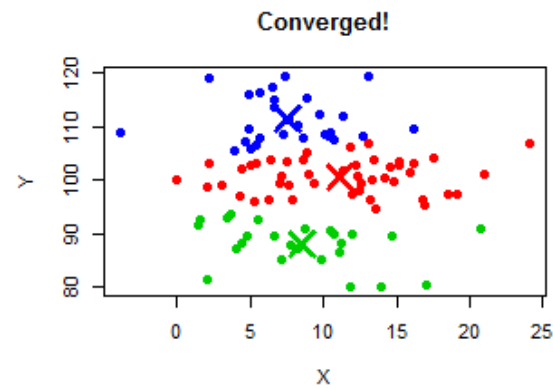
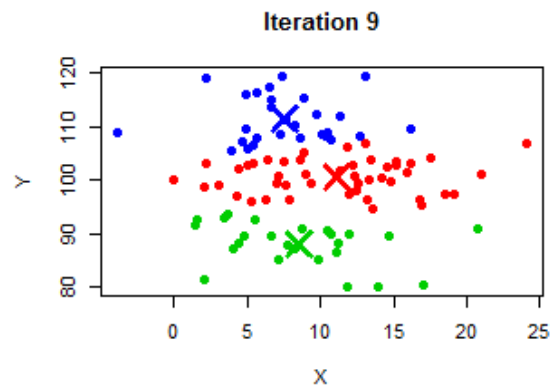
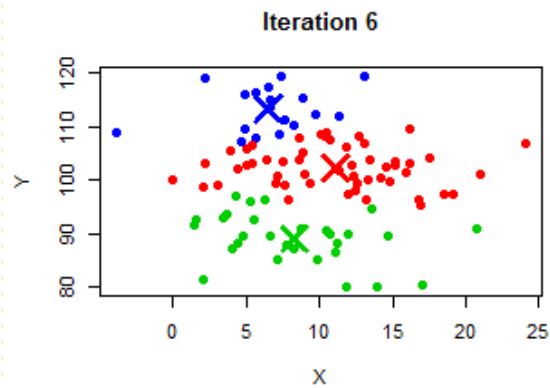
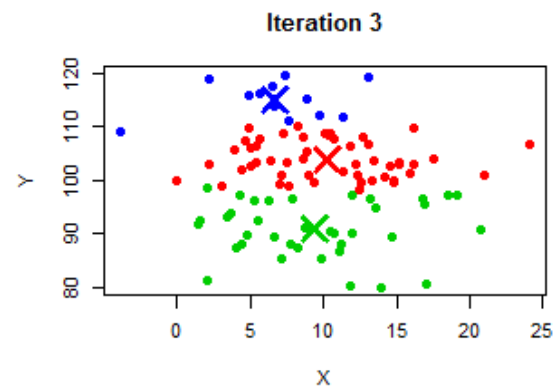
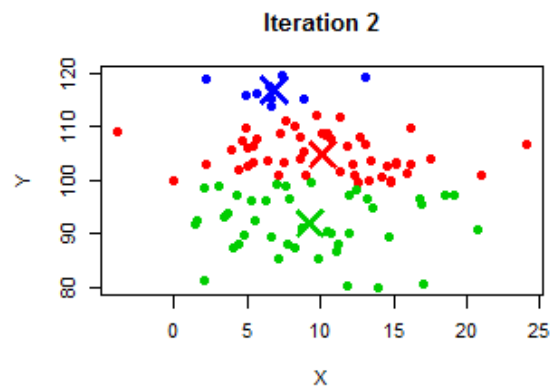
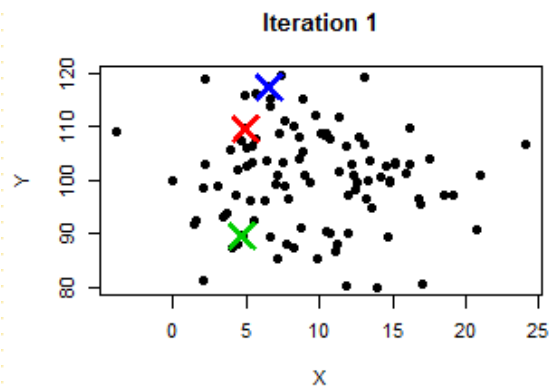
STEP 2: Calculate its distance from each cluster center

$$dist(x, c_i) = \sqrt{\sum_{j=1}^d (x_j - c_{ij})^2}$$



STEP 3: Recalculate the centers of each cluster

$$c_i = \frac{1}{|S_i|} \sum_{x \in S_i} x$$

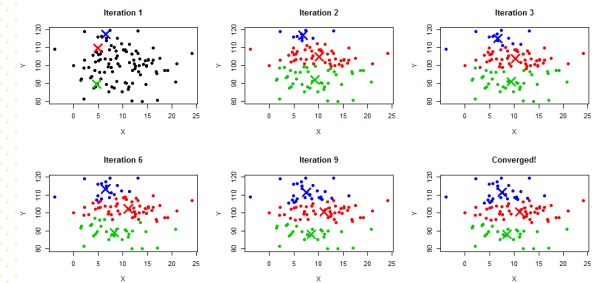


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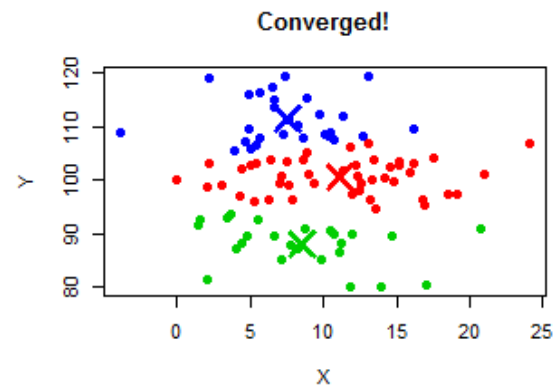
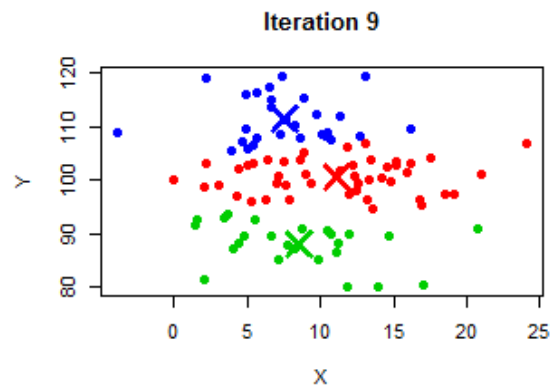
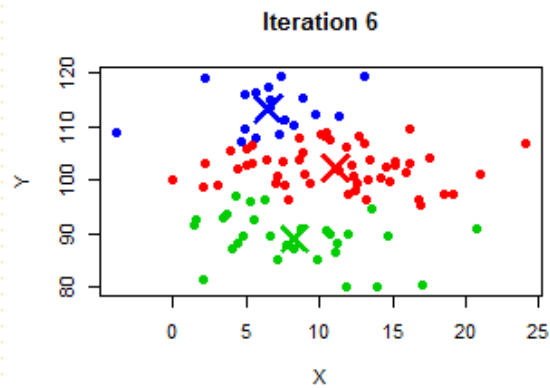
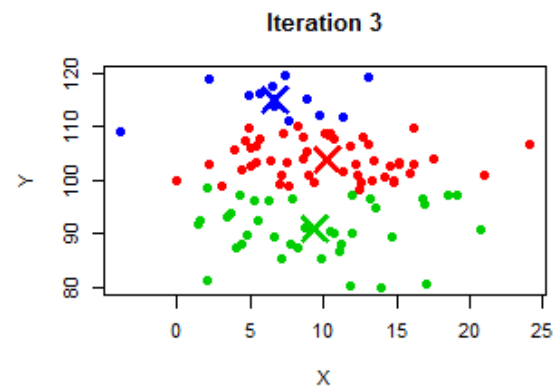
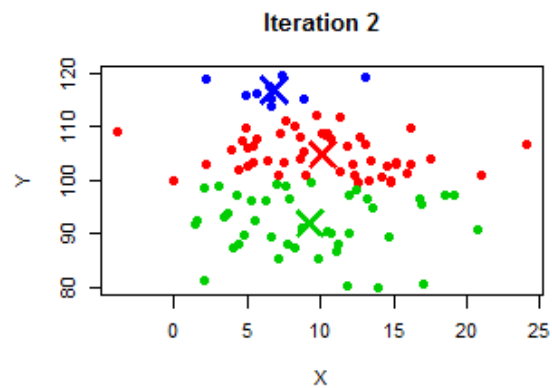
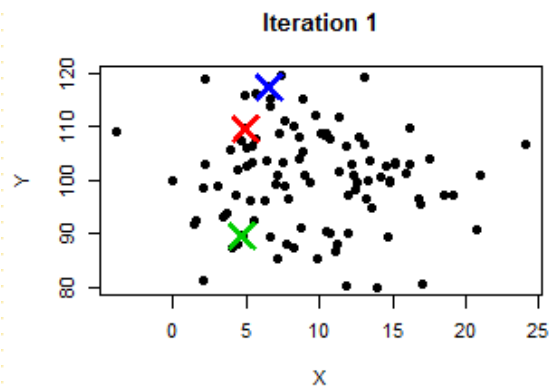
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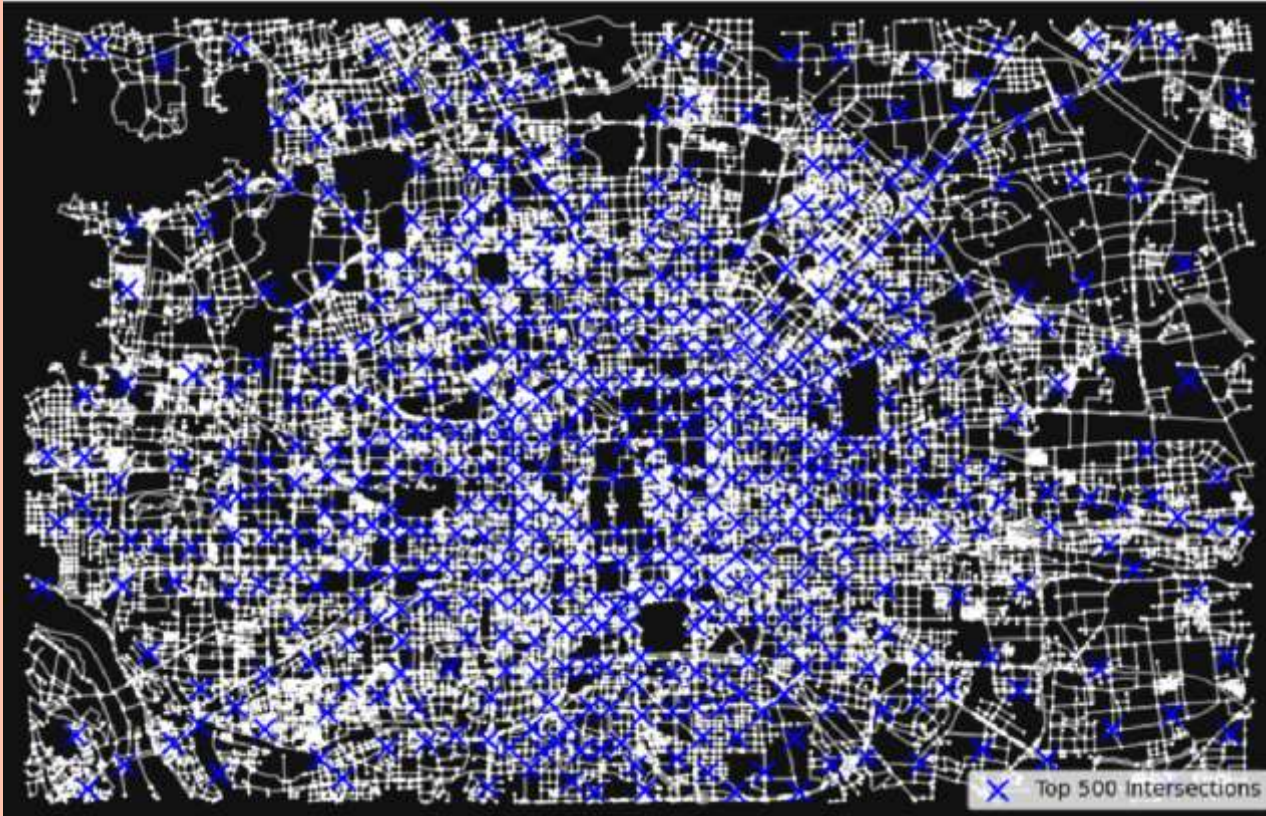
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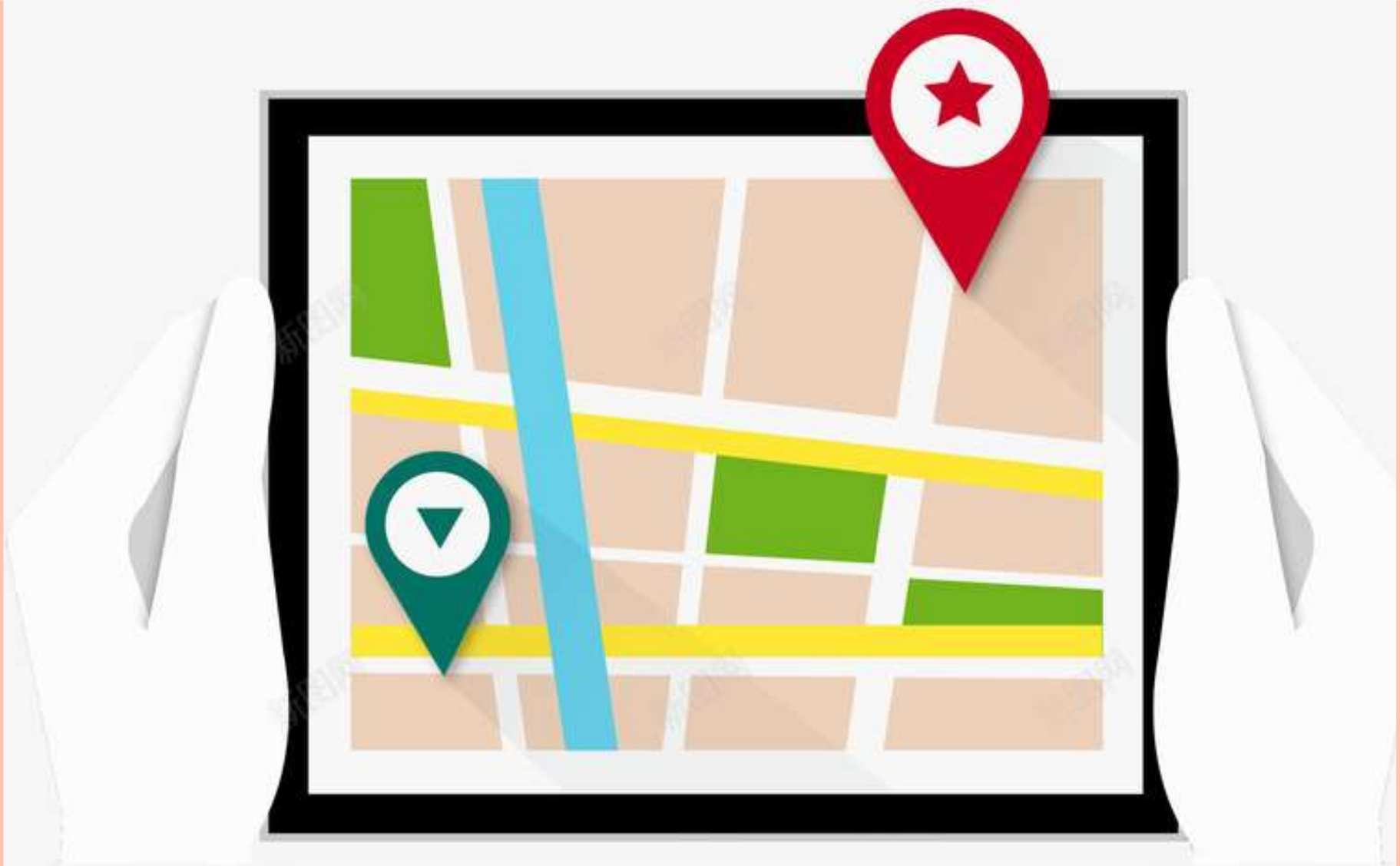


Maximum
Crossing
Points

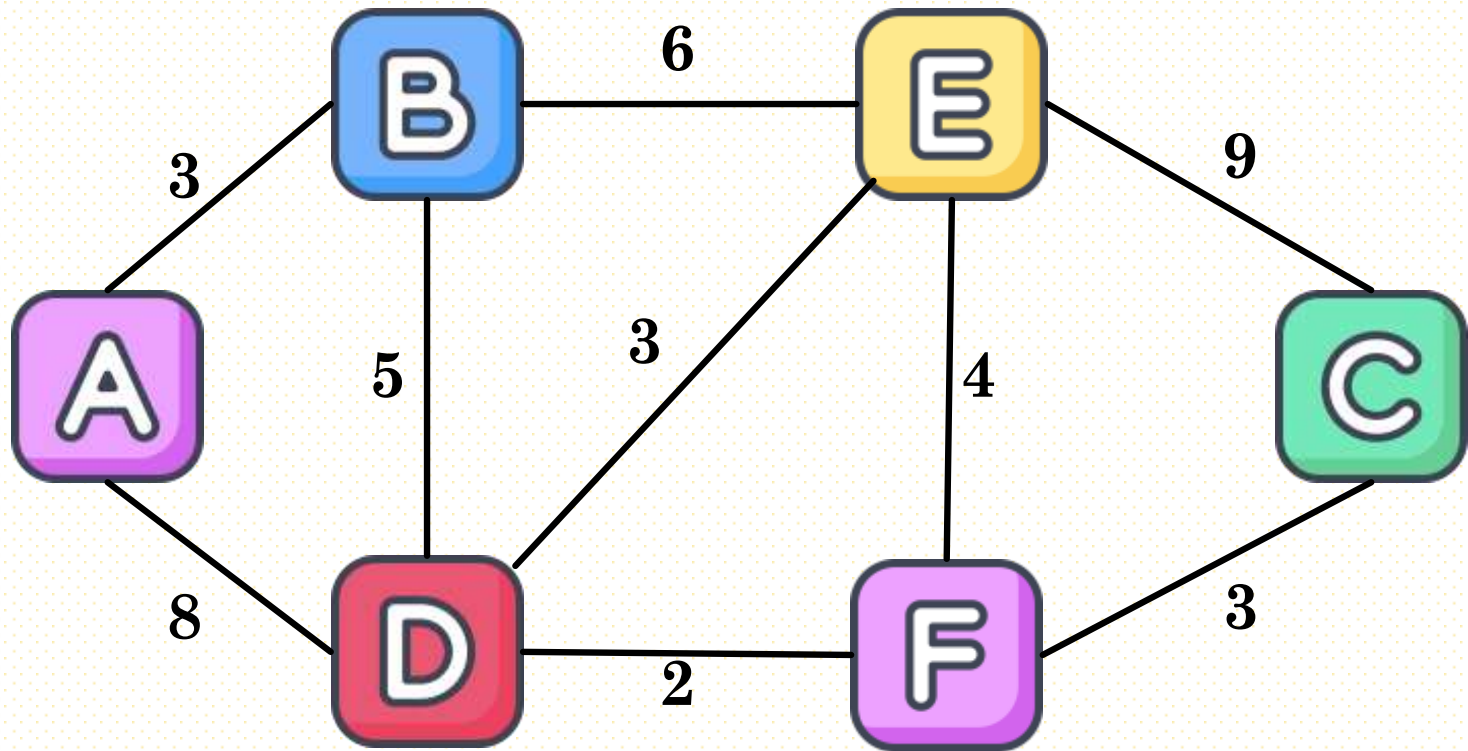


Landmark





Shortest path----Dijkstra algorithm



Distance:

A: 0

B: ~~3~~

C: ~~13~~

D: ~~8~~

E: ~~9~~

F: ~~10~~

Random
destination
starting
point

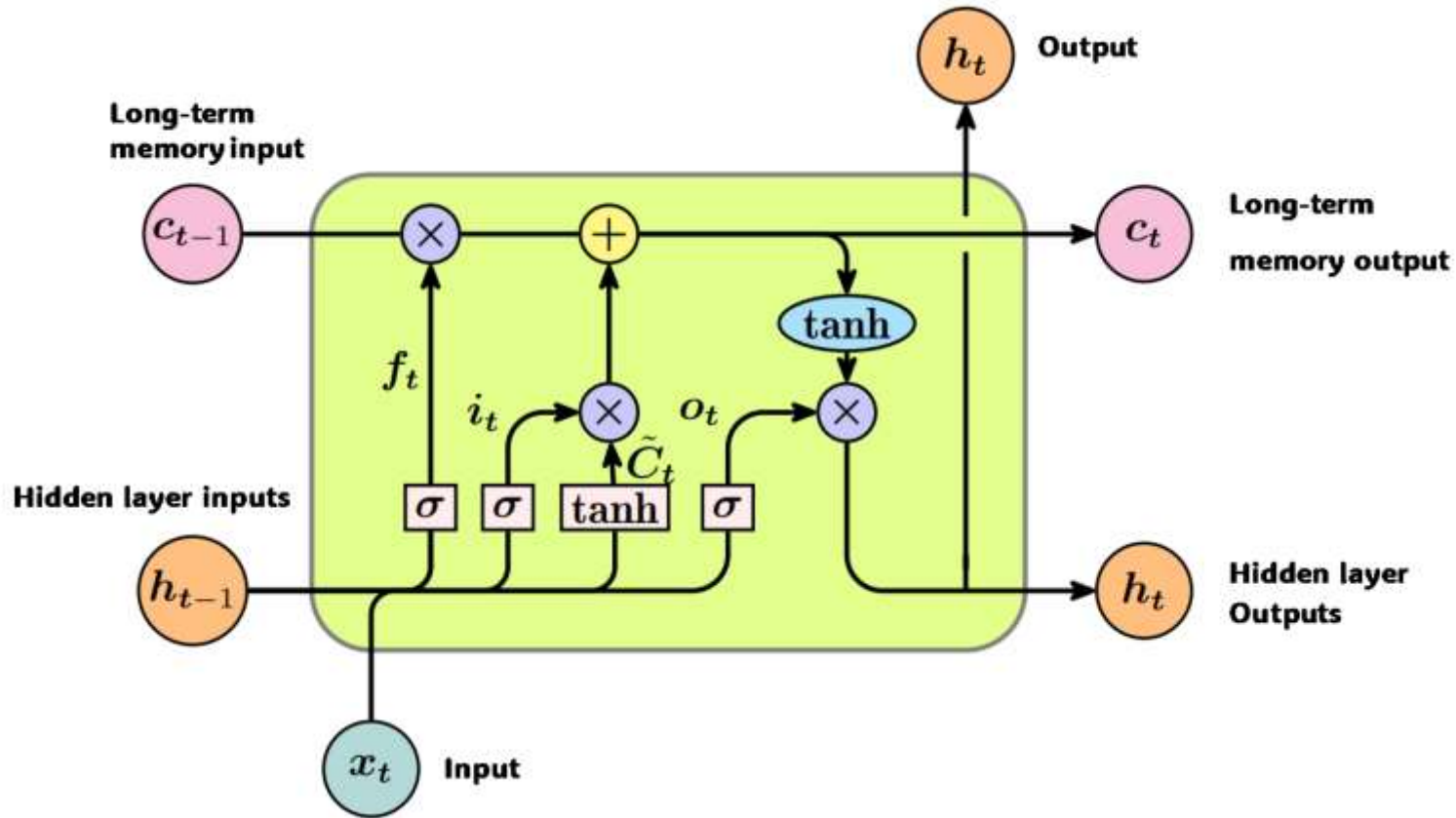


The
optimal
Path



TRAFFIC TRAJECTORY PREDICTION

TRAFFIC TRAJECTORY PREDICTION



EXPERIMENT

- **Target**

5 Consecutive Coordinate Points → Prediction Point

- **Experiment settings**

Normalization

◆ MinMaxScaler 0-1

LSTM layer

◆ 50 units

Training

◆ 5 epochs

Dataset Regulation

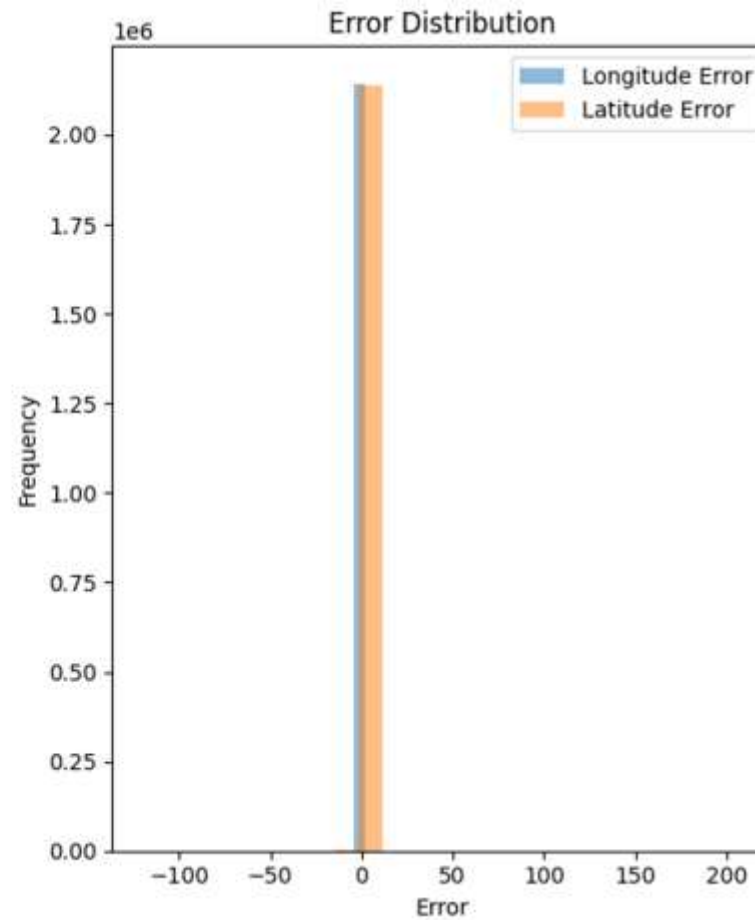
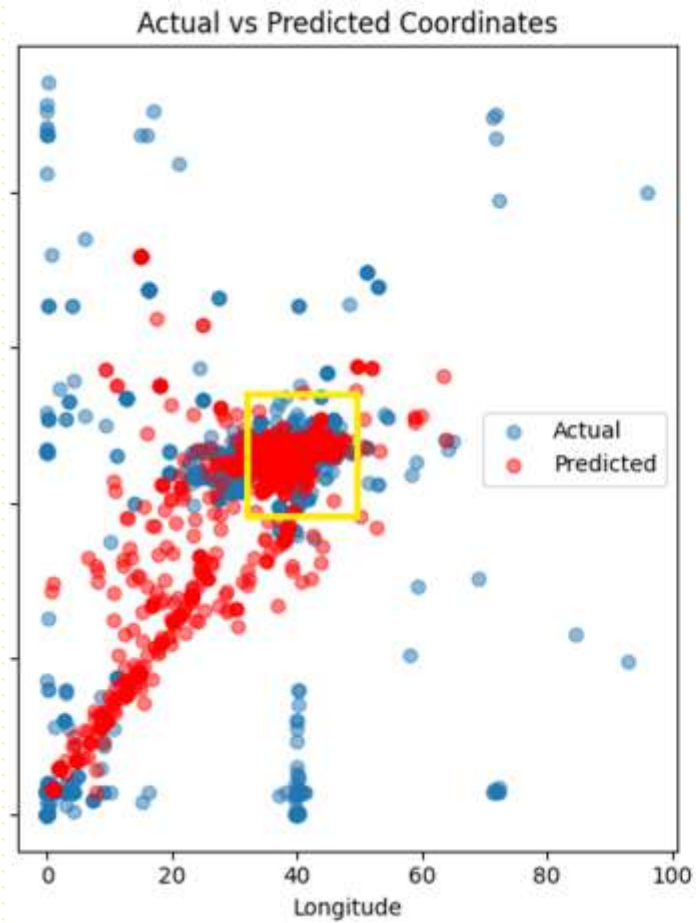
◆ 80% training & 20% testing

Evaluation Metrics

◆ MSE & RMSE

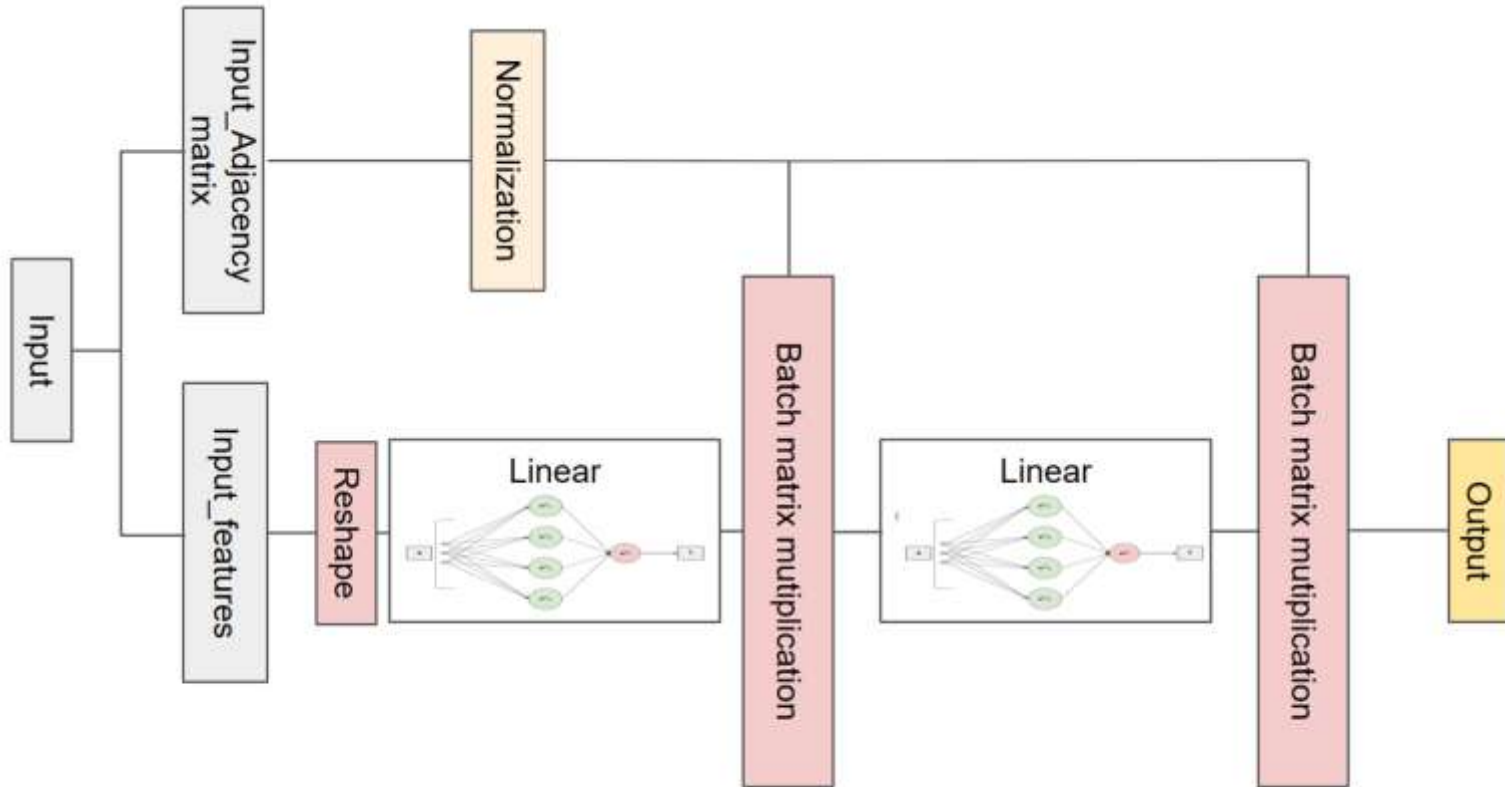
Evaluation Metrics	LSTM
MSE	1.98724
RMSE	1.40970

VISUALIZATION

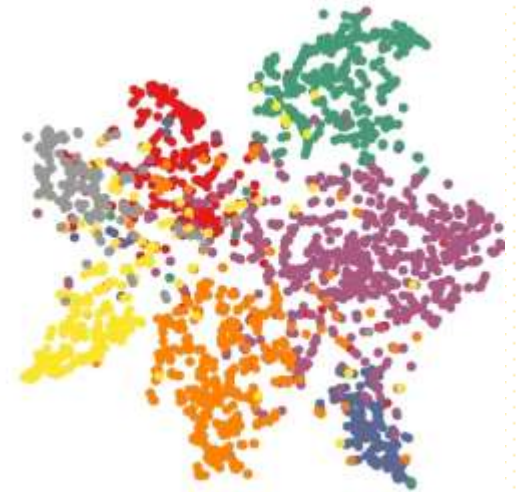
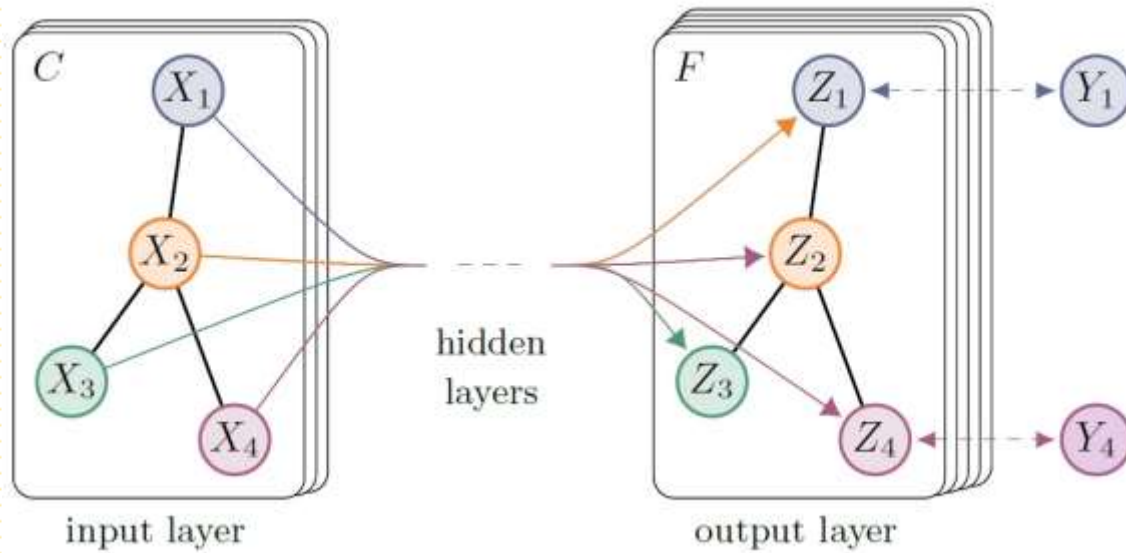


TRAFFIC FLOW PREDICTION

TRAFFIC FLOW PREDICTION



GCN



MLP

$$h^{(1)} = \sigma(W^{(0)}x + b)$$

The traditional MLP method can only use the historical coordinate information to speculate, and can not make full use of the spatial relationship between roads.

GCN

$$H^{(1)} = \sigma(D^{-1/2}A D^{-1/2}XW^{(0)})$$

GCN can make full use of these spatial information through graph convolution operation to predict the future moving position of the vehicle.

EXPERIMENT

- **Target**

Predicting the future traffic flow of a certain node

- **Experiment settings**

Learning rate

◆ 0.03

Training

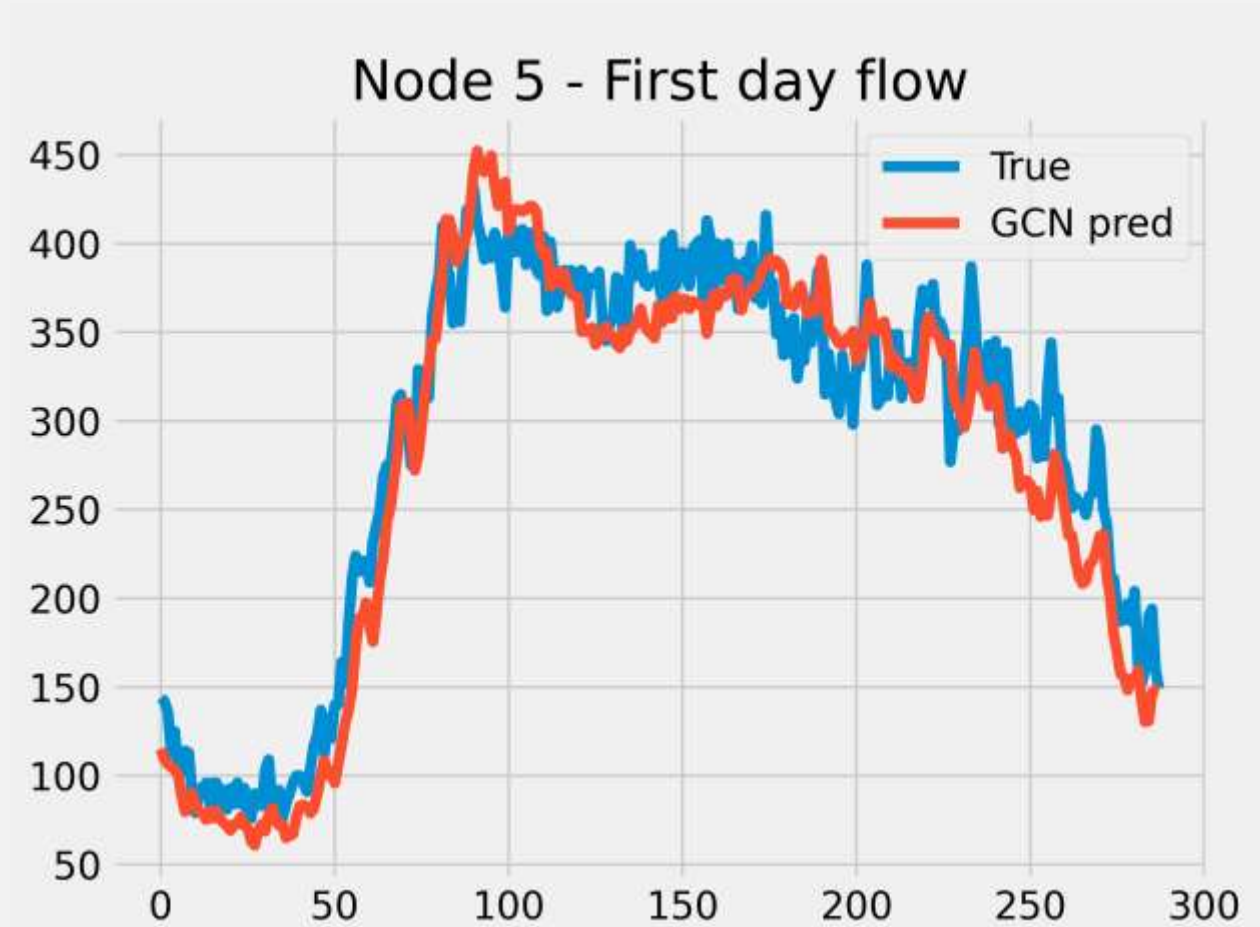
◆ 30 epochs

Evaluation Metrics

◆ MSE & RMSE & MAPE & TotalLoss

Evaluation Metrics	GCN
TotalLoss	0.0081
MSE	0.0645
RMSE	0.0900
MAPE	0.1279%

VISUALIZATION



DYNAMIC TIME WEIGHT ADJUSTMENT

$$w_{ij} = \frac{d_{ij}}{v_{ij}}$$

w_{ij} is the weight of the road segment from node i to node j.

d_{ij} is the length of the road segment.

v_{ij} is the average speed on that segment.

INNOVATION

◆ Multi model fusion: LSTM+GCN

- LSTM processes long-term dependencies in time series to predict changes in traffic flow.
- GCN aggregates adjacent road network nodes to improve path planning accuracy.

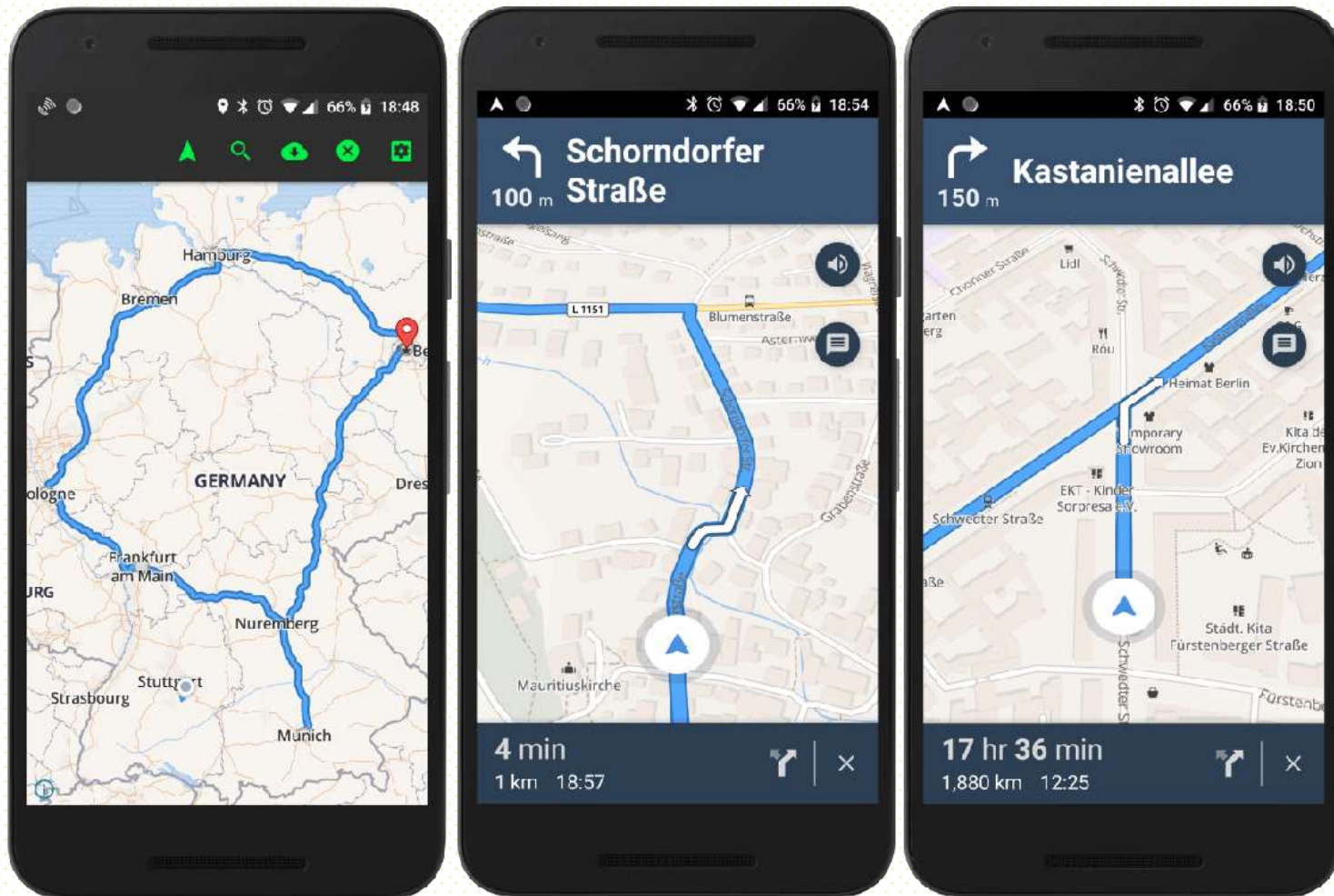
◆ Dynamic weight adjustment

- Real time response to traffic changes and dynamic optimization of path recommendations.

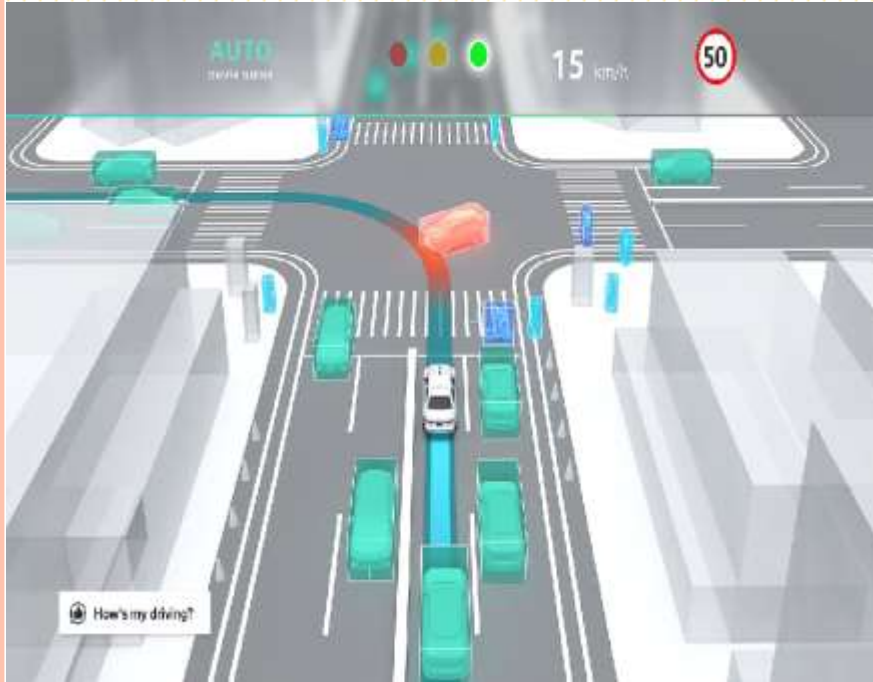
◆ Algorithm combination application

- Combining Dijkstra's algorithm to achieve optimal path planning in complex road networks.

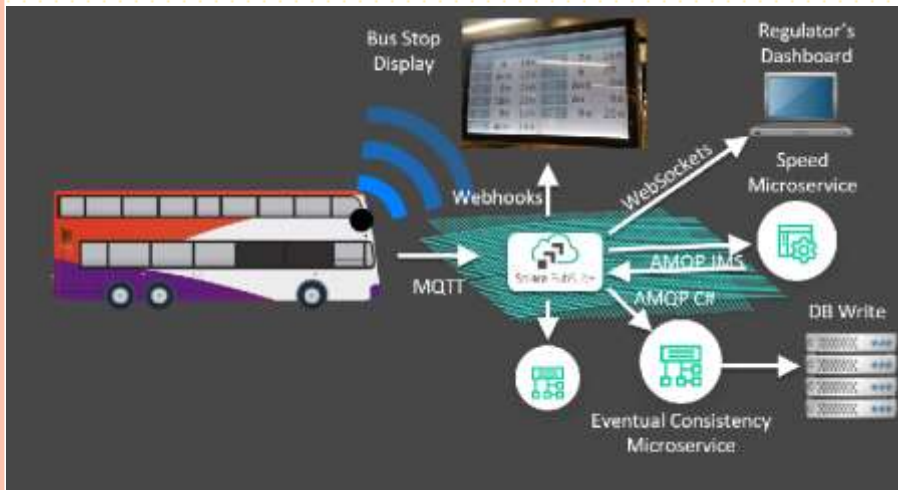
APPLICATION



AUTOMATED DRIVING



SMART CITY SIMULATION AND OPTIMIZATION



CONCLUSION

- ❑ Data Preparation
- ❑ Model Evaluation
- ❑ Innovation
- ❑ Model Application

THANK YOU